

COMPUTER SCIENCE MSC CURRICULUM

Program: Digital Factory Specialization

ASSESSMENT TYPES:

E: Exam | PG: Practice Grade | XE: Lecture+Practice with Exam | XPG: Lecture+Practice with Practical Grade

=== COMPULSORY SUBJECTS (74 Credits) ===

--- SEMESTER 1 (22 Credits) ---

IPMSEK-22RMEG: Research methodology

1 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | PG | 5 Credits

IPMSEK-23ldPLEGG: PLC programming and communications

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XPG | 5 Credits

IPMSEK-22ldMSLSG: Master studies and learning skills

0 Lecture, 3 Practice, 0 Laboratory, 0 Consultation | PG | 2 Credits

IPMSEK-22ldPAIEG: Principles of artificial intelligence

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits

IPMSEK-23ldPYEG: Python

1 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | XPG | 4 Credits

--- SEMESTER 2 (25 Credits) ---

IPMSEK-22ASTE: Advanced Software Technology (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPMSEK-22DAAE: Design and analysis of algorithms (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPMSEK-23ldERPSEG: ERP Systems

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits

IPMSEK-24ldIDEG: Industrial Databases

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPMSEK-22ldIMLDLEG: Introduction to Machine Learning and Deep Learning

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits

--- SEMESTER 3 (27 Credits) ---

IPMSEK-24ldPETEG: Privacy enhancing technologies

1 Lecture, 1 Practice, 0 Laboratory, 2 Consultation | XE | 4 Credits

IPMSEK-22ldDAMEG: Data Mining

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPMSEK-22ldDFLAB1: Digital Factory Lab I

0 Lecture, 0 Practice, 1 Laboratory, 4 Consultation | PG | 5 Credit

IPMSEK-22ldDFLAB2: Digital Factory Lab II

0 Lecture, 0 Practice, 1 Laboratory, 4 Consultation | PG | 5 Credits

IPMSEK-24ldROBEG: 3D technologies and Robotics

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XPG | 6 Credits

IPMSEK-22PRG: Internship

0 Credits | Semesters: 2-4 | 240 hours

=== COMPULSORY ELECTIVE SUBJECTS (Select 10 Credits) ===

--- SEMESTER 1 ELECTIVES (5 Credits) ---

IPMSEK-22fdHBPEG: Hardware based programming

1 Lecture, 3 Practice, 0 Laboratory, 1 Consultation | XPG | 5 Credits

IPMSEK-23fdTAMEG: Topics in Applied Mathematics

2 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | XPG | 5 Credits

--- SEMESTER 2 ELECTIVES (5 Credits) ---

IPMSEK-23fdAADEG: Advanced Application Design

2 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | XPG | 5 Credits

IPMSEK-22fdICEG: Industrial communications

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XPG | 5 Credits

=== THESIS & ADDITIONAL ===

IPMSEK-24THCONS: Thesis consultation
30 Credits | Semester: 4

IPMSEK-22ERASMUS: Erasmus mobility
Maximum 24 Credits | Semester: 3

Elective Subjects: 6 Credits | Semesters: 1,3

=== CREDIT SUMMARY ===

Compulsory Subjects: 74 Credits
Compulsory Elective Subjects: 10 Credits
Elective Subjects: 6 Credits
Thesis: 30 Credits

TOTAL DEGREE CREDITS: 120 Credits

Semester Credit Distribution:

Semester 1: 30 | Semester 2: 30 | Semester 3: 30 | Semester 4: 30